



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to CKB FUELS

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

#### Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

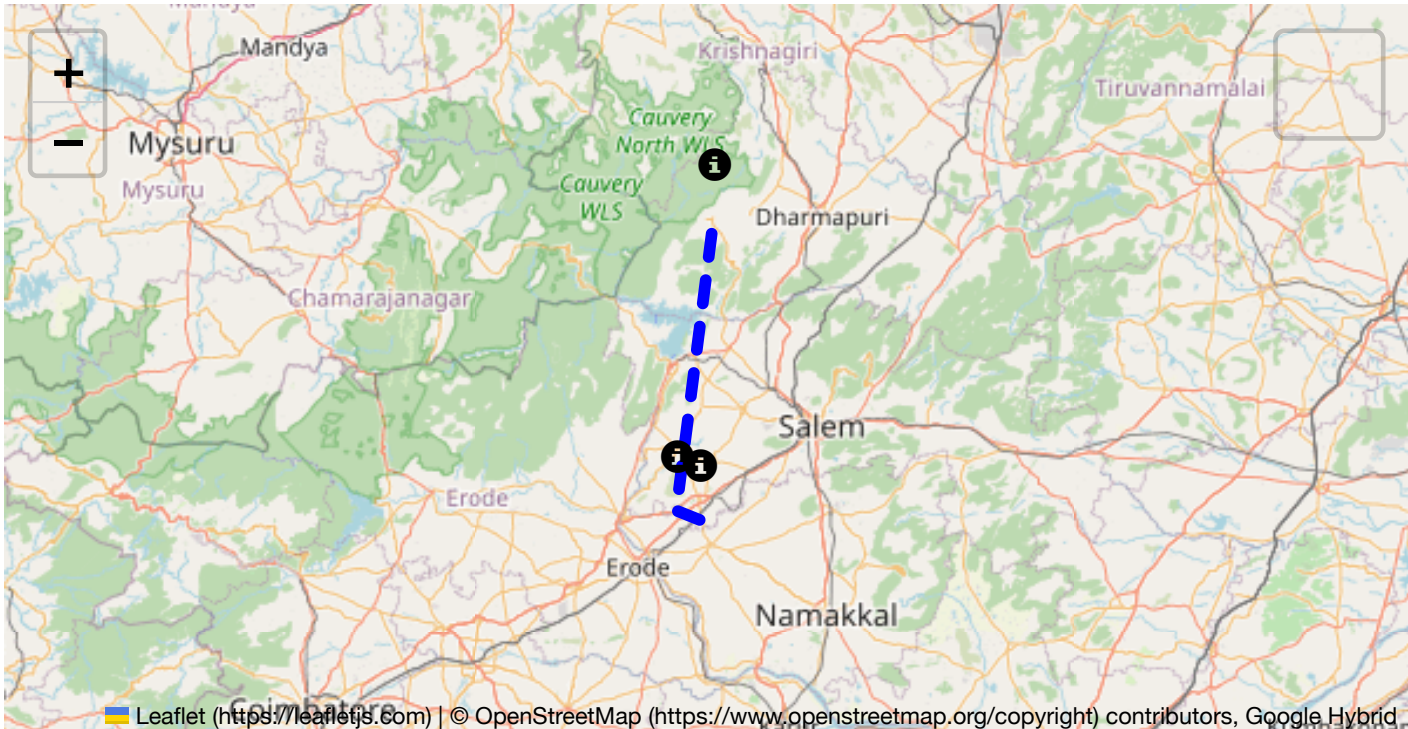
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

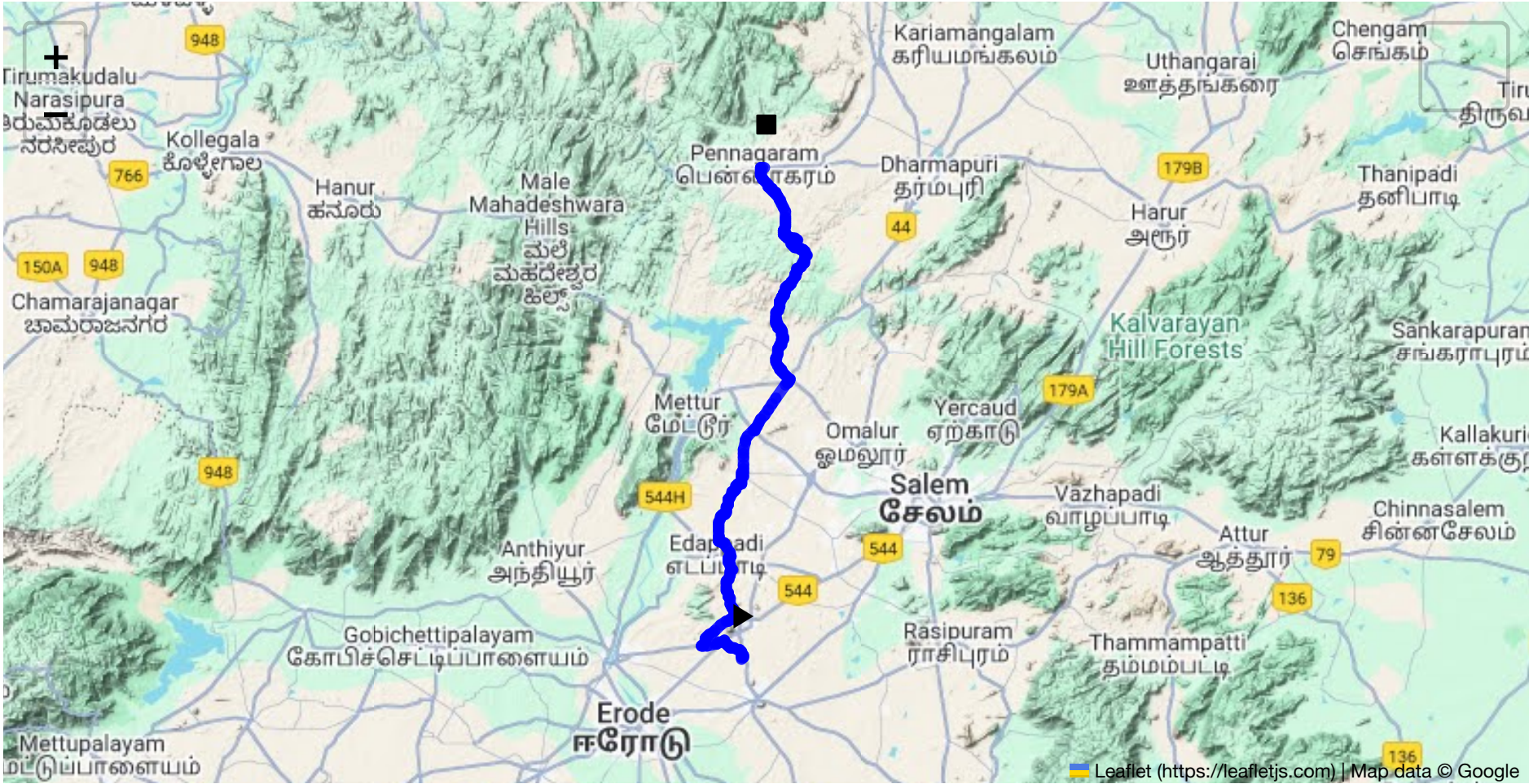
This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 103.76 km**  
**Estimated Duration: 2.5 hours**  
**Adjusted Duration (Heavy Vehicle): 3.2 hours**



Start: (11.4381, 77.8734)  
End: (12.131338, 77.906288)



## Welcome to the Journey Risk Management Study

To provide a comprehensive analysis of the route from CVQF+23W, Sangagiri to 4WJ4+JF, Kallipuram via Bhavani Main Road, here is a detailed examination of potential hazards and considerations for safer transport:

- 1. Overview of the Route Map:
  - The route is approximately 103.76 kilometers long and typically takes about 2.53 hours for heavy vehicles carrying hazardous materials, traveling generally in a northeast direction through a mix of rural and semi-urban areas.
- 2. Typical Weather Conditions and Potential Weather-Related Hazards:
  - Tamil Nadu generally experiences three main seasons: summer (around March to June), monsoon (July to November), and winter (December to February).
  - Heavy monsoon rains can lead to flooding, especially in low-lying areas and around river crossings which can impact road conditions and cause delays. Check weather forecasts regularly during the monsoon season for updates.
- 3. Analysis of Traffic Patterns:
  - Peak traffic hours typically occur from 8:00 AM to 10:00 AM and 5:00 PM to 8:00 PM due to local commuting.
  - Expect potential congestion near populated towns and junctions along the route, such as Saatyamangalam and suburban sections of Erode district.
- 4. Assessment of Road Quality and Infrastructure:
  - Bhavani Main Road is a key arterial road with variable quality. Some segments are well-maintained, but others, especially rural stretches, may have potholes or be less maintained.
  - Watch for road construction or repair activities which are common and might temporarily affect road quality.
- 5. Suggestions for Alternative Routes for Emergencies:

- In case of blockage or delays on Bhavani Main Road, consider detours via SH 15 or SH 21, which runs parallel and offers connectivity back to the main route.
- Utilize navigation tools in real-time to identify viable routes quickly.

6. **Summary of Local Regulations Affecting Hazardous Material Transport:**

- Tamil Nadu requires special permits for the transport of hazardous materials, and specific time frames might be prescribed for transit through certain areas to minimize risk to populated areas.
- Ensure compliance with all safety regulations and have documentation ready for inspection.

7. **Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:**

- Historical data is sparse, but occasional incidents have involved spillage and rollovers on provincial roads. Awareness and adherence to speed limits and road signs reduce risks.
- Report breakdowns or accidents quickly to local authorities to minimize delays and risks.

8. **Environmental Considerations and Sensitive Areas:**

- The route crosses several agricultural zones and occasional wildlife areas. Be mindful of local wildlife crossings and avoid spillage, which can severely affect sensitive ecosystems.
- Keep an eye out for signage indicating protected areas.

9. **Analysis of Communication Coverage:**

- Cellular network coverage is generally good along major roads but may be spotty in rural and forested areas. Pre-plan communication check-ins during low-coverage zone transits.

10. **Estimated Emergency Response Times:**

- Response times in urban areas like near Erode would likely be faster (around 15-30 minutes).
- In more rural stretches, response times might extend to 45 minutes or more depending on proximity to emergency services.

11. **Overall Summary of Risk Assessment:**

- The route is moderately challenging for hazardous material transport due to traffic congestion, varying road conditions, and environmental sensitivities.
- Proper planning, compliance with regulations, and real-time navigation adjustments are crucial in mitigating risks.

Overall, drivers should stay informed, practice defensive driving, and maintain communication with dispatch to ensure a safe journey through this route.

Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.58300, 77.85813 | 15 KM/Hr    |
| 1 | Roundabout | High       | 11.60348, 77.83967 | 15 KM/Hr    |
| 2 | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 3 | Turn       | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 4 | Turn       | High       | 11.44900, 77.87406 | 15 KM/Hr    |



|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 5  | Turn       | High       | 11.45353, 77.85697 | 15 KM/Hr    |
| 6  | Turn       | Medium     | 11.45781, 77.85203 | 30 KM/Hr    |
| 7  | Turn       | High       | 11.46307, 77.84941 | 15 KM/Hr    |
| 8  | Turn       | Medium     | 11.45546, 77.81800 | 30 KM/Hr    |
| 9  | Blind Spot | Blind Spot | 11.45605, 77.81683 | 10 KM/Hr    |
| 10 | Blind Spot | Blind Spot | 11.49397, 77.86740 | 10 KM/Hr    |
| 11 | Turn       | Medium     | 11.49441, 77.86690 | 30 KM/Hr    |
| 12 | Turn       | Medium     | 11.49971, 77.86468 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.52519, 77.85535 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.52808, 77.85576 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.54790, 77.85296 | 30 KM/Hr    |
| 16 | Turn       | Medium     | 11.55718, 77.85825 | 30 KM/Hr    |
| 17 | Turn       | High       | 11.57893, 77.85430 | 15 KM/Hr    |
| 18 | Turn       | Medium     | 11.57919, 77.85585 | 30 KM/Hr    |
| 19 | Turn       | Medium     | 11.58037, 77.85749 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.60978, 77.84165 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.61795, 77.84020 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.63085, 77.84072 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.64073, 77.84627 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.64130, 77.84816 | 30 KM/Hr    |
| 25 | Turn       | Medium     | 11.65274, 77.85197 | 30 KM/Hr    |
| 26 | Turn       | High       | 11.67310, 77.86031 | 15 KM/Hr    |
| 27 | Turn       | Medium     | 11.67357, 77.86193 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.68196, 77.86690 | 30 KM/Hr    |
| 29 | Turn       | High       | 11.68492, 77.86898 | 15 KM/Hr    |
| 30 | Turn       | High       | 11.68463, 77.87262 | 15 KM/Hr    |
| 31 | Turn       | High       | 11.69455, 77.87583 | 15 KM/Hr    |
| 32 | Turn       | High       | 11.69558, 77.87439 | 15 KM/Hr    |
| 33 | Turn       | High       | 11.76019, 77.89047 | 15 KM/Hr    |
| 34 | Turn       | High       | 11.75964, 77.89200 | 15 KM/Hr    |
| 35 | Turn       | Medium     | 11.83153, 77.94375 | 30 KM/Hr    |
| 36 | Turn       | High       | 11.83269, 77.94377 | 15 KM/Hr    |
| 37 | Turn       | Medium     | 11.84317, 77.93283 | 30 KM/Hr    |
| 38 | Turn       | Medium     | 11.84593, 77.92925 | 30 KM/Hr    |
| 39 | Turn       | Medium     | 11.84844, 77.92847 | 30 KM/Hr    |
| 40 | Turn       | Medium     | 11.88158, 77.93124 | 30 KM/Hr    |
| 41 | Turn       | Medium     | 11.88495, 77.93258 | 30 KM/Hr    |
| 42 | Turn       | Medium     | 11.88888, 77.93248 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 43 | Turn       | Medium     | 11.88979, 77.93330 | 30 KM/Hr    |
| 44 | Turn       | Medium     | 11.89093, 77.93364 | 30 KM/Hr    |
| 45 | Turn       | High       | 11.89818, 77.93069 | 15 KM/Hr    |
| 46 | Turn       | Medium     | 11.89893, 77.93090 | 30 KM/Hr    |
| 47 | Turn       | High       | 11.90023, 77.93247 | 15 KM/Hr    |
| 48 | Turn       | Medium     | 11.90434, 77.93007 | 30 KM/Hr    |
| 49 | Turn       | High       | 11.90535, 77.92665 | 15 KM/Hr    |
| 50 | Turn       | Medium     | 11.90979, 77.92570 | 30 KM/Hr    |
| 51 | Turn       | High       | 11.91187, 77.92304 | 15 KM/Hr    |
| 52 | Turn       | Medium     | 11.91408, 77.92286 | 30 KM/Hr    |
| 53 | Turn       | Medium     | 11.91463, 77.92308 | 30 KM/Hr    |
| 54 | Blind Spot | Blind Spot | 11.92989, 77.92663 | 10 KM/Hr    |
| 55 | Blind Spot | Blind Spot | 11.93022, 77.92591 | 10 KM/Hr    |
| 56 | Turn       | Medium     | 11.94153, 77.92988 | 30 KM/Hr    |
| 57 | Turn       | Medium     | 11.94365, 77.92965 | 30 KM/Hr    |
| 58 | Turn       | Medium     | 11.94962, 77.93394 | 30 KM/Hr    |
| 59 | Turn       | High       | 11.95109, 77.93458 | 15 KM/Hr    |
| 60 | Turn       | Medium     | 11.95167, 77.93416 | 30 KM/Hr    |
| 61 | Blind Spot | Blind Spot | 11.95462, 77.93515 | 10 KM/Hr    |
| 62 | Blind Spot | Blind Spot | 11.95363, 77.93517 | 10 KM/Hr    |
| 63 | Turn       | Medium     | 11.95431, 77.93560 | 30 KM/Hr    |
| 64 | Turn       | Medium     | 11.95642, 77.93570 | 30 KM/Hr    |
| 65 | Turn       | High       | 11.96346, 77.93881 | 15 KM/Hr    |
| 66 | Turn       | Medium     | 11.96404, 77.94036 | 30 KM/Hr    |
| 67 | Turn       | Medium     | 11.97772, 77.95193 | 30 KM/Hr    |
| 68 | Turn       | Medium     | 11.97993, 77.95175 | 30 KM/Hr    |
| 69 | Turn       | Medium     | 11.99874, 77.96519 | 30 KM/Hr    |
| 70 | Turn       | Medium     | 12.01168, 77.96562 | 30 KM/Hr    |
| 71 | Turn       | High       | 12.01606, 77.96149 | 15 KM/Hr    |
| 72 | Turn       | High       | 12.01756, 77.95626 | 15 KM/Hr    |
| 73 | Turn       | Medium     | 12.01699, 77.95316 | 30 KM/Hr    |
| 74 | Turn       | Medium     | 12.01845, 77.95149 | 30 KM/Hr    |
| 75 | Turn       | Medium     | 12.01900, 77.94967 | 30 KM/Hr    |
| 76 | Turn       | High       | 12.02410, 77.94698 | 15 KM/Hr    |
| 77 | Blind Spot | Blind Spot | 12.02402, 77.94615 | 10 KM/Hr    |
| 78 | Turn       | Medium     | 12.02473, 77.94638 | 30 KM/Hr    |
| 79 | Turn       | High       | 12.02552, 77.94751 | 15 KM/Hr    |
| 80 | Turn       | High       | 12.02664, 77.94739 | 15 KM/Hr    |

|     | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|-----|------------|------------|--------------------|-------------|
| 81  | Turn       | Medium     | 12.02760, 77.94981 | 30 KM/Hr    |
| 82  | Turn       | High       | 12.02840, 77.95098 | 15 KM/Hr    |
| 83  | Turn       | High       | 12.02826, 77.95162 | 15 KM/Hr    |
| 84  | Blind Spot | Blind Spot | 12.02959, 77.95304 | 10 KM/Hr    |
| 85  | Turn       | High       | 12.02941, 77.95259 | 15 KM/Hr    |
| 86  | Turn       | High       | 12.03000, 77.95158 | 15 KM/Hr    |
| 87  | Turn       | Medium     | 12.02942, 77.95021 | 30 KM/Hr    |
| 88  | Turn       | Medium     | 12.02966, 77.94883 | 30 KM/Hr    |
| 89  | Turn       | High       | 12.03115, 77.94769 | 15 KM/Hr    |
| 90  | Turn       | High       | 12.03047, 77.94496 | 15 KM/Hr    |
| 91  | Turn       | Medium     | 12.03113, 77.94425 | 30 KM/Hr    |
| 92  | Turn       | High       | 12.03172, 77.94417 | 15 KM/Hr    |
| 93  | Turn       | Medium     | 12.03212, 77.94335 | 30 KM/Hr    |
| 94  | Turn       | High       | 12.03211, 77.94102 | 15 KM/Hr    |
| 95  | Turn       | Medium     | 12.03275, 77.94040 | 30 KM/Hr    |
| 96  | Turn       | Medium     | 12.03333, 77.93760 | 30 KM/Hr    |
| 97  | Turn       | High       | 12.03567, 77.93479 | 15 KM/Hr    |
| 98  | Turn       | High       | 12.03946, 77.93582 | 15 KM/Hr    |
| 99  | Turn       | High       | 12.04055, 77.93376 | 15 KM/Hr    |
| 100 | Turn       | Medium     | 12.04125, 77.93379 | 30 KM/Hr    |
| 101 | Turn       | Medium     | 12.05056, 77.93669 | 30 KM/Hr    |
| 102 | Turn       | Medium     | 12.05186, 77.93623 | 30 KM/Hr    |
| 103 | Turn       | Medium     | 12.05638, 77.93704 | 30 KM/Hr    |
| 104 | Turn       | Medium     | 12.07449, 77.93475 | 30 KM/Hr    |
| 105 | Turn       | Medium     | 12.09326, 77.92357 | 30 KM/Hr    |
| 106 | Turn       | High       | 12.09460, 77.92352 | 15 KM/Hr    |
| 107 | Turn       | High       | 12.10263, 77.91301 | 15 KM/Hr    |
| 108 | Turn       | Medium     | 12.10489, 77.91289 | 30 KM/Hr    |
| 109 | Turn       | Medium     | 12.12839, 77.90114 | 30 KM/Hr    |
| 110 | Turn       | Medium     | 12.13163, 77.90491 | 30 KM/Hr    |

Risk Assessment - Elevation Changes

|   | Risk Type        | Specific Risk | Risk Level | Coordinates                          | Speed Limit |
|---|------------------|---------------|------------|--------------------------------------|-------------|
| 0 | Elevation Change | Ascent        | Medium     | 12.03000891961515, 77.94856304281852 | 30 KMS/Hr   |

Emergency Locations

|    | type     | name                                           | coordinates               | speed_limit | risk_level |
|----|----------|------------------------------------------------|---------------------------|-------------|------------|
| 1  | hospital | Parusuraman Hospital                           | 11.697164,<br>77.877792   | 30 km/h     | Medium     |
| 2  | hospital | JP Nature Cure Hospital Siddha and Naturopathy | 11.7002949,<br>77.8770897 | 30 km/h     | Medium     |
| 3  | hospital | SSS Hospital                                   | 11.7501471,<br>77.8832809 | 30 km/h     | Medium     |
| 4  | clinic   | DSR Clinic                                     | 11.7603176,<br>77.8908853 | 30 km/h     | Medium     |
| 5  | hospital | Srilakshmi Hospital                            | 11.7601, 77.892272        | 30 km/h     | Medium     |
| 6  | hospital | Government Hospital                            | 11.758158,<br>77.894528   | 30 km/h     | Medium     |
| 8  | hospital | Vasu Speciality Hospital                       | 11.8331097,<br>77.943169  | 30 km/h     | Medium     |
| 9  | hospital | Rajamanickam Hospital                          | 11.8332408,<br>77.9431081 | 30 km/h     | Medium     |
| 10 | hospital | Child Hospital                                 | 11.833001,<br>77.943415   | 30 km/h     | Medium     |
| 11 | hospital | Naveen Nursing Home                            | 11.8335355,<br>77.9437441 | 30 km/h     | Medium     |
| 12 | clinic   | Sri Krishna Clinic                             | 11.835443,<br>77.9450587  | 30 km/h     | Medium     |
| 13 | hospital | Rishan Hospital                                | 11.879358,<br>77.9286133  | 30 km/h     | Medium     |
| 14 | hospital | Government Hospital Kadaimadai                 | 12.05434, 77.936888       | 30 km/h     | Medium     |
| 15 | hospital | Government Hospital, Pennagaram                | 12.129564,<br>77.9041522  | 30 km/h     | Medium     |

Crowded Spots

|   | type    | name                            | coordinates            | speed_limit | risk_level |
|---|---------|---------------------------------|------------------------|-------------|------------|
| 0 | school  | KRP Matric. Hr. Sec School      | 11.4546193, 77.8142445 | 30 km/h     | Medium     |
| 7 | college | Kavery Educational Institutions | 11.8255319, 77.9342256 | 30 km/h     | Medium     |



## Route Photos of Risky Spots



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.58300, 77.85813



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.60348, 77.83967



**Risk Type:** Turn

**Risk Level:** High



**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.44029, 77.87544**



**Risk Type: Turn**  
**Risk Level: High**  
**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.44900, 77.87406**



**Risk Type: Turn**  
**Risk Level: High**  
**Speed Limit: 15 KM/Hr**  
**Coordinates: 11.45353, 77.85697**





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45781, 77.85203



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.46307, 77.84941



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45546, 77.81800





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.45605, 77.81683



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.49397, 77.86740



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.49441, 77.86690





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.49971, 77.86468



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.52519, 77.85535



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.52808, 77.85576





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.54790, 77.85296



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.55718, 77.85825



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.57893, 77.85430





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.57919, 77.85585



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.58037, 77.85749



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.60978, 77.84165





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.61795, 77.84020



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.63085, 77.84072



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64073, 77.84627





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64130, 77.84816



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.65274, 77.85197



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.67310, 77.86031





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.67357, 77.86193



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.68196, 77.86690



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.68492, 77.86898





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.68463, 77.87262



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.69455, 77.87583



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.69558, 77.87439





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.76019, 77.89047



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.75964, 77.89200



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.83153, 77.94375

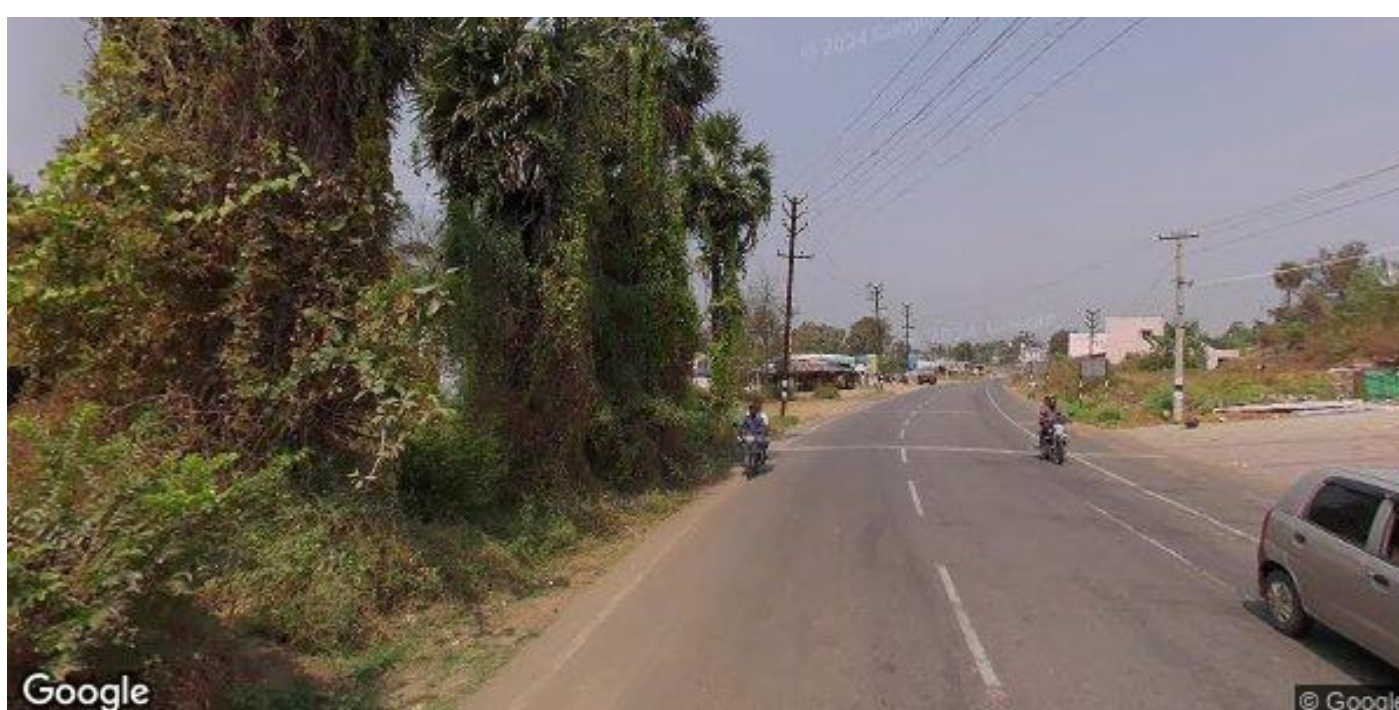




**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.83269, 77.94377



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.84317, 77.93283



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.84593, 77.92925





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.84844, 77.92847



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.88158, 77.93124



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.88495, 77.93258





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.88888, 77.93248



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.88979, 77.93330



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.89093, 77.93364





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.89818, 77.93069



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.89893, 77.93090



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.90023, 77.93247





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.90434, 77.93007



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.90535, 77.92665



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.90979, 77.92570





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.91187, 77.92304



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.91408, 77.92286



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.91463, 77.92308





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.92989, 77.92663



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.93022, 77.92591



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.94153, 77.92988





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.94365, 77.92965



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.94962, 77.93394



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.95109, 77.93458





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95167, 77.93416



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.95462, 77.93515



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.95363, 77.93517





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95431, 77.93560



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95642, 77.93570



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96346, 77.93881





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96404, 77.94036



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.97772, 77.95193



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.97993, 77.95175





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.99874, 77.96519



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.01168, 77.96562



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.01606, 77.96149





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.01756, 77.95626



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.01699, 77.95316



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.01845, 77.95149





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.01900, 77.94967



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.02410, 77.94698



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 12.02402, 77.94615





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.02473, 77.94638



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.02552, 77.94751



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.02664, 77.94739





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.02760, 77.94981



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.02826, 77.95162



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 12.02959, 77.95304





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.02941, 77.95259



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.03000, 77.95158



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.02942, 77.95021





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.02966, 77.94883



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.03115, 77.94769



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.03047, 77.94496





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.03113, 77.94425



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.03172, 77.94417



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.03212, 77.94335





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.03211, 77.94102



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.03275, 77.94040



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.03333, 77.93760





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.03567, 77.93479



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.03946, 77.93582



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.04055, 77.93376





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.04125, 77.93379



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.05056, 77.93669



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.05186, 77.93623





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.05638, 77.93704



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.07449, 77.93475



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.09326, 77.92357





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.09460, 77.92352



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.10263, 77.91301



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.10489, 77.91289





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.12839, 77.90114



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.13163, 77.90491



